

etc), it becomes necessary to check whether the app is consuming additional memory. An app that works efficiently with fewer resources is considered effective.

Lab testing: Laboratory testing provides the mobile app with a simulated wireless environment without actual network coverage. It is implemented in a way that the coverage and behaviour of calls, SMSs and other functions can be evaluated on different bandwidths.

Load testing: This analyses how the mobile app performs when exposed to many users at one time. Load testing checks whether the app crashes or hangs when there is a heavy load of users. For example, the mobile app is opened concurrently by one thousand users to check whether it is able to handle such a load.

Location testing: This includes testing whether the app is able to identify the network coverage and location correctly. The compatibility of the app with different networks is checked during this test.

Installation testing: This process checks whether the application can be installed on all types of devices with different configurations. For example, the app can be tested to be installable on different versions of the mobile operating system and on handsets with different memory constraints.

Penetration testing: The process of penetration testing checks whether the forms and online widgets of the app are fully secured. This is always required in the case of sensitive and financial apps like those used for e-commerce, online shopping, banking, etc. The hackers should not be able to penetrate the payment gateway of the app with their tricks.



Figure 1: Testing strategies for mobile applications



Figure 2: Home page of the Appium automation testing tool

Open source tools for testing Android smartphone apps

The three open source tools listed in this article provide different functions that rigorously analyse the behaviour of the mobile app when exposed to diverse traffic.

Appium

URL: <http://appium.io/>

Appium is a powerful free and open source automation testing tool for native, mobile and hybrid apps. It provides cross-platform features to execute the tests on multiple mobile handsets.

Appium can be used for the testing of Android, iOS or Windows mobile apps in association with the WebDriver protocol. It supports multiple browsers, including Chrome and Safari, for testing apps.

Assets	
Appium-1.6.0-mac.zip	199 MB
Appium-1.6.1.dmg	189 MB
Appium-1.6.1.dmg.blockmap	203 MB
appium-desktop-1.6.1-full.nupkg	141 MB
appium-desktop-1.6.1-ia32.nsis.7z	110 MB
appium-desktop-1.6.1-mac.zip	197 MB
appium-desktop-1.6.1-x86_64.AppImage	126 MB
appium-desktop-Setup-1.6.1-ia32.exe	142 MB
appium-desktop-setup-1.6.1.exe	110 MB
appium-desktop-setup-1.6.1.exe.blockmap	108 MB
appium-desktop-web-setup-1.6.1.exe	696 KB
latest-linux.yml	394 Bytes
latest-mac.json	177 Bytes
latest-mac.yml	480 Bytes
latest.yml	668 Bytes

Figure 3: Downloading Appium Desktop

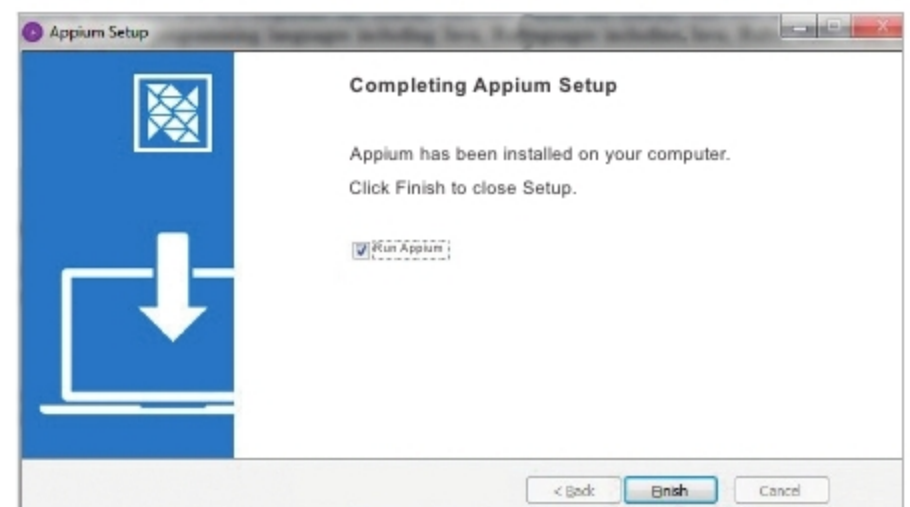


Figure 4: Running Appium

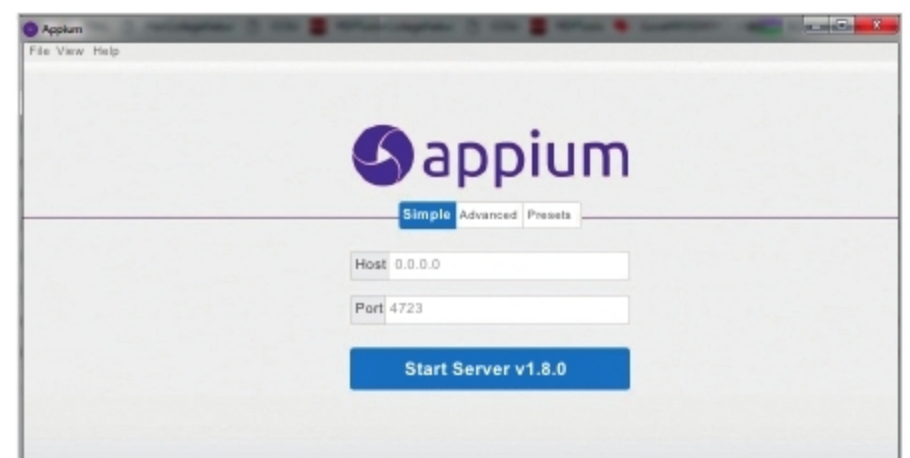


Figure 5: Starting Appium Server